8a	Fast moving/ high speed/ high (kinetic) energy electrons moving towards the target metal (anode), undergo rapid/large deceleration/ acceleration as they interact with the lattice ions. The loss in energy of the electrons is emitted in the form of X-ray photons, and since the loss of energy can vary according to the interaction, they form the continuous X-ray spectrum.	**B1** **B1** **B1**
8b	tungsten is ductile so it can be made into thin wires tungsten has a high atomic number tungsten is a good thermionic emitter / good at emitting electrons tungsten has a very high melting point (any one relevant point)	**A1**
8ciA.	From Fig. 8.2, μ of muscle is 0.40 cm^{-1}. Using $\frac{I}{I_0} = e^{-\mu x}$, ratio along MM = $e^{-0.40(8.0)}$ = 0.041 or 0.0408	**C1** **A1**
8ciB.	From Fig. 8.2, μ of bone is 2.46 cm^{-1}. Using $\frac{I}{I_0} = e^{-\mu x}$, ratio along BB = $\left(\frac{I}{I_0}\right)_{muscle} \times \left(\frac{I}{I_0}\right)_{bone}$ $= e^{-0.40(4.0)} \times e^{-2.46(4.0)}$ = 0.000011 or 0.0000108 or ratio along BB = $\left(\frac{I}{I_0}\right)_{muscle,\ 2\ cm} \times \left(\frac{I}{I_0}\right)_{bone} \times \left(\frac{I}{I_0}\right)_{muscle,\ 2\ cm}$ $= e^{-0.40(2.0)} \times e^{-2.46(4.0)} \times e^{-0.40(2.0)}$ = 0.000011 or 0.0000108	**C1** **A1**
8cii	Since the ratio along BB is 3 orders of magnitude smaller than that along MM, the contrast is good. Accept qualitative answer - $\frac{0.0408}{1.08\times10^{-5}}$=3793 (B1)	**B1** **A1**
8d	• small space, not suitable for patient who are claustrophobic • strong magnetic field can cause ferromagnetic objects in the room to become projectile posing risk to patient and staff • metal implants (e.g. pacemaker, cochlear implants) / ferromagnetic implants in patients may malfunction / move in the strong magnetic field	**A1**

	• (long scan time) patient needs to remain still for long duration and any reasonable challenges e.g. any movement may cause blurred image, children or patients who cannot lie down for long duration. • ineffective for bone imaging due to fast relaxation times • helium gas leaking during a quench resulting in suffocation or frostbite (any one relevant point)	
8e	For a solenoid, $B = \mu_0 nI$ $1.5 = (4\pi\times10^{-7}) \times n \times 750$ $n = 1591.55\ m^{-1}$	**C1** **A1**
8f	Lamor frequency, $\omega = \gamma B_0$ $= (2.68\times10^{8}) \times 1.5$ $= 4.02\times10^{8}\ rad\ s^{-1}$	**C1** **A1**
8g	frequency of RF wave = $\omega/2\pi$ $= 4.02\times10^{8}/2\pi$ $= 64$ or $64.0\ MHz$	**C1** **A1**
8h	assuming all electrical energy is used to vaporise liquid helium and constant current of 750 A, rate of electrical heating = rate of vaporisation of liquid helium $I^2R = ml_v$ where m is the initial rate of vaporization $(750)^2(0.0045) = m\ (21\times10^{3})$ $m = 0.12$ or $0.121\ kg\ s^{-1}$	**C1** **A1**